

Data Cleaning & Visualization Project

– Salary Dataset

Tools: Python, Pandas, Matplotlib, Seaborn

Dataset: Salary_Data.xlsx (Age, Gender, Education Level, Job Title, Years of Experience, Salary)

1. Objective

Clean a raw salary dataset (missing values, duplicates, outliers, inconsistent labels) and visualize the key insights in a dashboard.

2. Data Cleaning Summary

Step	Issue found	Action	Rows affected
Column names	Hidden BOM character in first column (İ»çAge)	Renamed columns	–
Empty rows	2 fully blank rows	Dropped	2

Step	Issue found	Action	Rows affected
Duplicates	Same record repeated many times	<code>drop_duplicates()</code>	4,911
Missing values	Salary missing (target column)	Dropped rows (never guess the target)	3
Missing values	Education Level (1)	Filled with mode	1
Missing values	Years of Experience (1)	Filled with median	1
Inconsistent labels	Bachelor's / Bachelor's Degree , Master's / Master's Degree , PhD / PhD	Standardized	~1,050
Inconsistent labels	Social M , Social Media Man	Changed to Social Media Manager	2
Outliers – Salary	Salaries of 350–579 (data-entry errors; IQR can't catch them because	Removed (Salary < 1000)	4

Step	Issue found	Action	Rows affected
	its lower bound is negative)		
Outliers – Experience	1 person aged 25 with 12 years experience (impossible)	Removed	1
Outliers – IQR check	Age: 6 flagged, Experience: 22 flagged, Salary: 0 flagged	Kept – these are genuine senior employees	0

Result: 6,704 raw rows → 1,783 clean rows (about 73% of the raw file was duplicate data).

3. Key Findings

1. Experience is the strongest driver of salary (correlation 0.82). Average pay grows from about \$49K (0–2 years) to about \$182K (20+ years).
2. Education matters a lot: High School ≈ \$45K, Bachelor's ≈ \$92K, Master's ≈ \$127K, PhD ≈ \$161K.
3. Overall pay: average salary is \$113,399, median \$110,000, maximum \$250,000. The distribution is fairly flat with a slight tail toward high earners.
4. Top-paying roles are leadership positions: Director of Data Science (\$200K), ~~Marketing Director~~ (\$176K), Software Engineer

Manager (~\$173K).

5. Gender gap: male average (\$118K) is about 10% higher than female average (\$108K). This is a raw comparison and does not account for experience or job role, so it should not be read as proof of unequal pay.
6. Age and experience are almost the same signal (correlation 0.94), so use only one of them if you build a prediction model.

4. Deliverables

- `Dashboard.png` – 5 KPI cards and 6 charts in one view
- `charts/` – 8 individual charts (including a before/after outlier boxplot)
- `Salary_Data_Cleaned.csv` – cleaned dataset
- `cleaning_log.csv` – row counts for every cleaning step
- `salary_analysis.py` – complete, re-runnable code

5. Conclusion

After cleaning, the data shows that experience, education level and seniority of the role explain most of the salary differences. Removing the duplicates was the biggest fix, since they would otherwise have skewed every average.